Independently of the functional form, the CERN wage return in Eq. (3) sums up to return to education (EDU), experience (EXP),[20] and other training beyond CERN (T).

An unbiased OLS estimation requires ε_i to have a zero mean and to be uncorrelated with CERN, a condition met only if participation in CERN training is random. However, training participation is influenced by factors such as past academic performance, ambition, and unobserved individual abilities, which also affect wages. We control for observable factors that may jointly influence training duration and salary formation. Firstly, we account for observable confounding factors by estimating an augmented model (Eq. 5):

$$\ln wage_i = \beta_0 + \beta_1 EDU_i + \beta_2 f(EXP_i) + \beta_3 f(CERN) + \beta_4 \bar{S}_i + \beta_5 \bar{Z}_i + \beta_6 \bar{C} + \varepsilon_i \qquad (5)$$

where $\bar{S}_i$, $\bar{Z}_i$, and $\bar{C}$ are vectors of salary-related information, individual traits, and contextual variables, such as country of employment and sector.[21] Subsequently, to test H1b and H2, the earnings function is further augmented to include skills acquired or improved during the CERN training as in Eq. (6):

$$\begin{aligned}\ln wage_i = {} & \beta_0 + \beta_1 EDU_i + \beta_2 f(EXP_i) \\ & + \beta_3 f(CERN) + \beta_4 f(Skills)_{ij} + \beta_5 \bar{S}_i + \beta_6 \bar{Z}_i + \beta_7 \bar{C} + \varepsilon_i \qquad (6)\end{aligned}$$

where $f(Skills)_{ij}$ represents different functional forms, including the number of skills acquired or a binary variable indicating whether at least one skill was acquired. The subscript j denotes the skill type (hard or soft). Following Camporesi et al. (2017) and Catalano et al. (2021), we grouped scientific and technical skills, critical analysis, problem-solving capacity, and independent thinking under the label "hard skills", while communication, team/project leadership, developing maintaining and using networks of collaborations under the label "soft skills" (see Section 4.2.1 for details). Section 4.2.3 also proposes a set of robustness estimates that partially account for ability bias in Eq. (5) and jointly control for it in Eq. (6).

[20] Despite usually considered in its quadratic form, when the working experience horizon refers to the first years of experience, it can be considered linear experience (Sørensen and Vejlin 2014).
[21] $\bar{C}$ is not individual-dependent and therefore the subscript i is omitted.

4. Analysis and results

4.1 Descriptive analysis of the targeted sample

ECRs in our selected sample were from 91 different research institutions worldwide, most from Europe. Respondents were predominantly male, with 86% (n = 168) reflecting the male-dominated nature of the HEP research field (Holman et al. 2018). The age of respondents ranged from a minimum of 24 years (1%; n = 2) to a maximum of 40 years (3%; n = 6). Specifically, 59% (n = 116) were under 35 years old, and an additional 15% were 38 years old (Table B.1, Annex B).

The majority of ECRs (61%; n = 119) held or were pursuing a doctoral degree (Table 4), which is consistent with the high average number of years of formal education in our sample (19.8 years). More than half of the sample (51%; n = 111) had attended additional training courses alongside their formal education and CERN research training (Table B.1, Annex B).

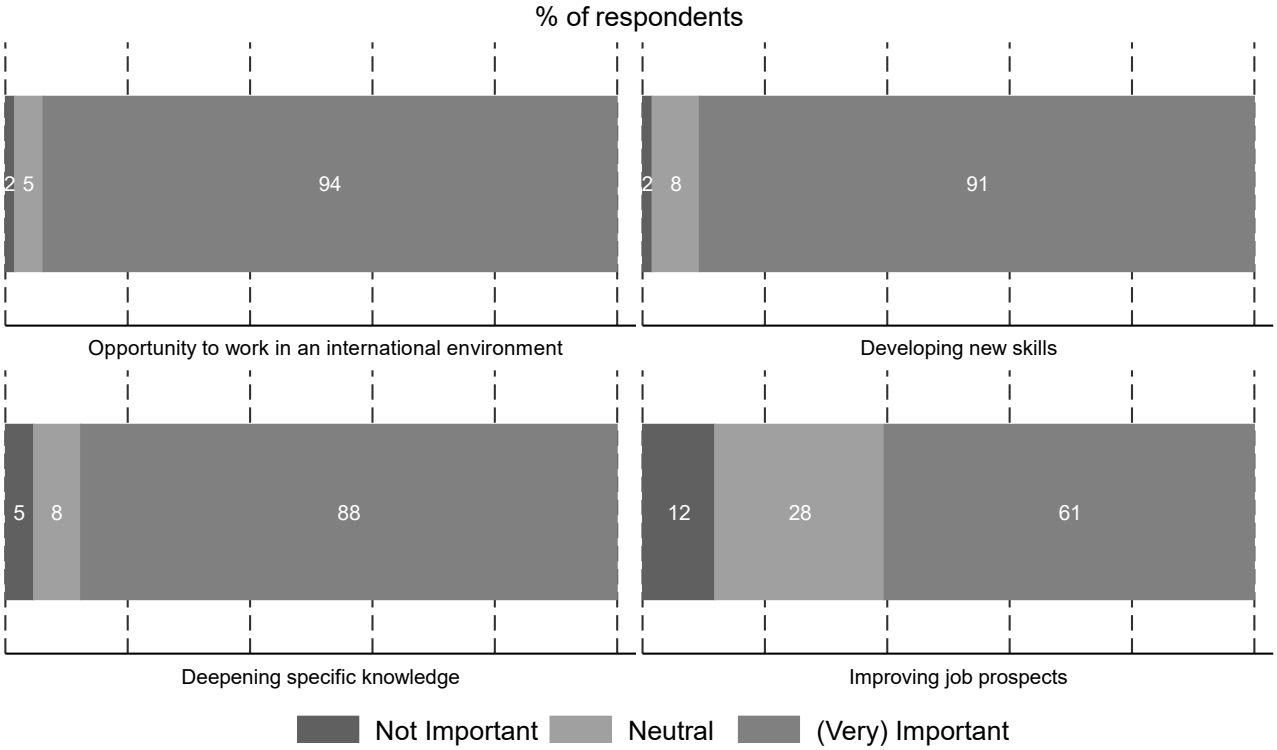
Table 4 – Education level of respondents (*N = 196*)

Education level	N	(%)
Doctoral degree	112	57%
Doctoral degree ongoing (at the time of the survey)	6	3%
Higher professional education	6	3%
Master degree	62	32%
University degree (below master's)	9	5%
Secondary diploma	1	1%

Note: Rounded figures. Source: authors' elaboration

Through the survey, we investigated the research training experience at CERN from multiple perspectives, posing a series of questions that addressed various dimensions of the training process. These included the decision to apply, the duration of the training, the nature of the research activities undertaken, and the skills acquired or enhanced throughout the research experience. To gain a deeper understanding of the factors driving their decision to apply, we presented ECRs with a set of options and asked them to rank them according to their importance. The opportunity to work in an international environment and develop new skills emerged as the primary motivations for applying to CERN, with over 90% of respondents categorising these factors as "Important" or "Very Important." The desire to deepen specific knowledge ranked next, with 88% of respondents acknowledging its significance. In contrast, improving job prospects was cited by 60% of ECRs, while 12% considered it either "Not at all important" or "Low importance."

Figure 1. Motivation to apply to a research training experience at CERN (*N = 196*)



Source: authors' elaboration. Note*:* 196 valid answers analysed. Multiple answers question. Question C.4 of the questionnaire: "How do you rate the importance of the following considerations on your decision to apply for a working experience at CERN? Respondents were asked to indicate their answer on a Likert scale including 5 options (Not at all important, Low importance, Neutral, Important, Very important). In the chart, the option "(Very) important" groups the original options "Very Important" and "Important" while "Not Important" groups the original options "Low importance" and "Not at all important".

The duration of active period at CERN lasted between 1 and 5 years for 86% (*n = 169*) of them, with a peak of 2 years. The average duration in the sample is 3.4 years (Figure 2).

ECRs were mainly engaged in experimental physics (*32%; n = 63*), engineering (*28%; n = 54*), particle accelerator physics (*14%; n = 28*), and informatics (*13%; n = 26*) research domains. The remaining part carried out research or other activities in different domains, mostly including theoretical physics, management and administration (e.g. marketing, health and safety, human resources) (Figure 3).